

Unit 40: Integration of Rational Functions by Partial Fractions

1. The idea, in plain words

You already know how to combine fractions like $\frac{1}{x-1} + \frac{1}{x+1}$ into a single fraction: $\frac{2x}{x^2-1}$. Partial fractions is that exact process in reverse — taking one complicated fraction and splitting it back apart into simple pieces that are each easy to integrate on their own.

Why bother? A fraction like $\frac{2x+6}{(x+1)(x-1)^2}$ has no obvious antiderivative as written. But if you can rewrite it as $\frac{A}{x+1} + \frac{B}{x-1} + \frac{C}{(x-1)^2}$ for the right constants A, B, C , then suddenly you're just integrating three separate, simple pieces — each one either a basic $\ln$ form or a basic power rule form.

Before doing anything else, always check one thing first: is the numerator's degree at least as large as the denominator's? If so, you can't do partial fractions directly — you have to do polynomial long division first, just like dividing $17 \div 5 = 3$ remainder 2. This splits your fraction into a plain polynomial part (easy to integrate directly) plus a genuinely smaller “leftover” fraction — and it's only that leftover piece that needs partial fractions.

A valuable shortcut worth watching for: sometimes the numerator is exactly the derivative of the denominator (or a constant multiple of it). When that happens, you don't need partial fractions at all — a simple substitution gives you a clean $\ln$ answer immediately. Always glance for this pattern before diving into the full decomposition process.

Setting up the decomposition:

- Distinct linear factors, like $(x-a)(x-b)$: split into $\frac{A}{x-a} + \frac{B}{x-b}$.
- A repeated linear factor, like $(x-a)^2$: you need both $\frac{A}{x-a}$ and $\frac{B}{(x-a)^2}$ — not just one term.

Finding the constants — the fast way: multiply both sides of your decomposition equation by the full original denominator to clear all fractions. Then, plug in the value of x that makes each factor zero, one at a time — this instantly kills off all the other terms and hands you each constant directly.

2. Toolbox

Step 0 (always check first): if $\deg(\text{numerator}) \geq \deg(\text{denominator})$, perform polynomial long division before anything else. Only the remainder fraction gets decomposed.

Setting up partial fractions:

$$\frac{P(x)}{(x-a)(x-b)} = \frac{A}{x-a} + \frac{B}{x-b} \qquad \frac{P(x)}{(x-a)^2(x-b)} = \frac{A}{x-a} + \frac{B}{(x-a)^2} + \frac{C}{x-b}$$

Solving for constants: clear the denominators, then plug in each root of the denominator one at a time (the “cover-up” trick) — or, if a constant can't be isolated that way (common with repeated factors), plug in one more convenient value (like $x=0$) and solve the small remaining system.

Integrating each piece:

$$\int \frac{A}{x-a} dx = A \ln|x-a| + C \quad \int \frac{A}{(x-a)^2} dx = -\frac{A}{x-a} + C$$

The shortcut to watch for: if the numerator equals (or is a constant multiple of) the derivative of the denominator, skip decomposition entirely:

$$\int \frac{f'(x)}{f(x)} dx = \ln|f(x)| + C$$

3. Common mistakes

- Forgetting to do polynomial long division first when the numerator's degree is at least the denominator's degree — partial fractions only works correctly on a “proper” fraction (numerator degree strictly less than denominator degree).
- Setting up a repeated factor incorrectly, using only $\frac{B}{(x-a)^2}$ and forgetting the accompanying $\frac{A}{x-a}$ term.
- Arithmetic slips solving for the constants. Double-check each one by plugging back into the cleared-denominator equation.
- Forgetting the absolute value in $\ln|x-a|$.
- Missing the derivative-matching shortcut, and doing a full (unnecessary) partial fraction decomposition when a quick substitution would have worked.
- Sign errors when clearing denominators or distributing, especially with three or more factors — write out each step rather than trying to do it in your head.

4. Problem Set

Warm-up

1. $\int \frac{dx}{(x-1)(x+1)}$
2. $\int \frac{dx}{(x-2)(x+3)}$
3. $\int \frac{3x+5}{(x+1)(x+2)} dx$
4. $\int \frac{5x-1}{(x-1)(x-3)} dx$
5. $\int \frac{dx}{x^2-4}$
6. $\int \frac{x+7}{x^2+x-6} dx$ (factor the denominator first!)

Standard

7. $\int \frac{dx}{(x-1)^2(x+1)}$
8. $\int \frac{3x-1}{x^2(x+1)} dx$

9. $\int \frac{dx}{(x-1)(x-2)(x-3)}$
10. $\int \frac{2x+1}{(x-2)^2(x+1)} dx$
11. Use polynomial long division to rewrite $\frac{x^3+2x^2-5}{x+3}$ as a polynomial plus a remainder fraction.
12. $\int \frac{x^2+1}{x-1} dx$ (long division needed first!)

Challenge

13. $\int \frac{2x-6}{x^2-6x+8} dx$
14. $\int \frac{2x-7}{x^2-7x+12} dx$
15. $\int \frac{2x+6}{(x+1)(x-1)^2} dx$
16. $\int \frac{x+1}{(x-1)(x-2)(x-3)} dx$
17. $\int \frac{x^3+3x^2+2x-7}{x+2} dx$

Applied

18. Find $\int \frac{dx}{(x-2)(x-5)}$ (this type of integral arises when solving certain reaction-rate differential equations).
19. Find $\int \frac{dP}{P(100-P)}$ (this is a key building block for solving the logistic growth equation).
20. Find $\int_0^2 \frac{x+3}{(x+1)(x+2)} dx$.
21. Find $\int_4^5 \frac{dx}{(x-3)(x-6)}$.